YOTSUBA SHADER

Backend compatibility

Which graphics backends support each concept of the language, verified with their shader toolchains

Edition 1.0 · October 2026 · Data: tools/compat/compat.json of the website repository

A shader written with Yotsuba Shader is compiled to SPIR-V and translated to the format of each graphics backend. Not every backend can express every concept: WebGPU has no render target layer, WebGL 2 has no compute stage, Metal has no 64-bit floats. This document lists, concept by concept, what each backend supports. It is the rule set that the IDE uses to warn you while you write a shader.

How to read it

Status	Meaning
Yes	The concept works. At least one translator produced output that the backend's own compiler accepts, without extensions.
Extension	It works only with an extension or an optional feature, named in the details. The IDE reports a warning.
Lost	The output compiles, but the translator silently removed the concept. Treated as unsupported.
Invalid	The translator produced output, but the backend's compiler rejects it.
No	No translator can express the concept for this backend.
No stage	The backend has no such shader stage (for example, compute on WebGL 2).

Compatibility by concept

Concept	Vulkan (SPIR-V)	Direct3D 12 (HLSL SM6)	Metal (MSL)	WebGPU (WGSL)	OpenGL 4.5 (GLSL)	WebGL 2 (GLSL ES 3.00)
Vertex stage built-ins						
Clip-space position (vertex output) `Position`	Yes	Yes	Yes	Yes	Yes	Yes
Vertex index `VertexIndex`	Yes	Yes	Yes	Yes	Yes	Yes
Instance index `InstanceIndex`	Yes	Yes	Yes	Yes	Yes	Yes
Base vertex `BaseVertex`	Yes	Yes	Yes	No	Extension	Invalid
Base instance `BaseInstance`	Yes	Yes	Yes	No	Extension	Invalid
Clip distance `ClipDistance`	Yes	Yes	Yes	Extension	Yes	Extension
Render target layer `Layer`	Yes	Yes	Yes	No	Extension	Invalid

Concept	Vulkan (SPIR-V)	Direct3D 12 (HLSL SM6)	Metal (MSL)	WebGPU (WGSL)	OpenGL 4.5 (GLSL)	WebGL 2 (GLSL ES 3.00)
Viewport index `ViewportIndex`	Yes	Yes	Yes	No	Invalid	Invalid
Fragment stage built-ins						
Fragment coordinate `FragCoord`	Yes	Yes	Yes	Yes	Yes	Yes
Primitive index `PrimitiveId`	Yes	Yes	Yes	Extension	Yes	Invalid
Front-facing flag `FrontFacing`	Yes	Yes	Yes	Yes	Yes	Yes
Fragment depth (output) `FragDepth`	Yes	Yes	Yes	Yes	Yes	Yes
Stencil reference (output) `FragStencilRef`	Extension	No	Yes	No	Extension	No
Sample index `SampleId`	Yes	Yes	Yes	Yes	Yes	Invalid
Sample mask `SampleMask`	Yes	Yes	Yes	No	Yes	Invalid
Sample position `SamplePosition`	Yes	No	Yes	No	Yes	Invalid
Helper invocation `HelperInvocation`	Yes	Yes	Yes	No	Yes	Invalid
Barycentric coordinates `BaryCoord`	Extension	Yes	Yes	Invalid	Extension	Invalid
Barycentric coordinates (no perspective) `BaryCoordNoPerspective`	Extension	Yes	Yes	Invalid	Extension	Invalid
Conservative depth `ConservativeDepth`	Yes	Yes	Yes	Lost	Yes	Lost
Compute stage						
Global invocation id `GlobalInvocationId`	Yes	Yes	Yes	Yes	Yes	No stage
Local invocation id `LocalInvocationId`	Yes	Yes	Yes	Yes	Yes	No stage
Local invocation index `LocalInvocationIndex`	Yes	Yes	Yes	Yes	Yes	No stage
Workgroup id `WorkgroupId`	Yes	Yes	Yes	Yes	Yes	No stage
Number of workgroups `NumWorkgroups`	Yes	Yes	Yes	Yes	Yes	No stage
Workgroup size (built-in value) `WorkgroupSize`	Yes	Yes	Yes	Yes	Yes	No stage
Workgroup size (method attribute) `LocalSize`	Yes	Yes	Yes	Yes	Yes	No stage
Subgroups						

Concept	Vulkan (SPIR-V)	Direct3D 12 (HLSL SM6)	Metal (MSL)	WebGPU (WGSL)	OpenGL 4.5 (GLSL)	WebGL 2 (GLSL ES 3.00)
Subgroup size `SubgroupSize`	Yes	Yes	Yes	Extension	Extension	No stage
Subgroup invocation id `SubgroupInvocationId`	Yes	Yes	Yes	Extension	Extension	No stage
Subgroup id `SubgroupId`	Yes	Yes	Yes	Extension	Extension	No stage
Number of subgroups `NumSubgroups`	Yes	Yes	Yes	Extension	Extension	No stage
Interface						
Interface location `Location`	Yes	Yes	Yes	Yes	Yes	Yes
Resources						
Uniform buffer `UniformBuffer`	Yes	Yes	Yes	Yes	Yes	Yes
Read-only storage buffer `StorageBufferReadOnly`	Yes	Yes	Yes	Yes	Yes	Invalid
Read/write storage buffer `StorageBufferReadWrite`	Yes	Yes	Yes	Yes	Yes	Invalid
Sampled texture and sampler `TextureSampler`	Yes	Yes	Yes	Yes	Yes	Yes
Explicit set and binding `DescriptorSetBinding`	Yes	Yes	Yes	Yes	Yes	Yes
Types						
64-bit float (double) `Double`	Yes	Yes	No	Invalid	Yes	No
Stages						
Vertex stage `VertexStage`	Yes	Yes	Yes	Yes	Yes	Yes
Fragment stage `FragmentStage`	Yes	Yes	Yes	Yes	Yes	Yes
Compute stage `ComputeStage`	Yes	Yes	Yes	Yes	Yes	No stage

Details of every limitation

Concept	Backend	Status	Reason (tools)
Base vertex	WebGPU (WGSL)	No	**naga:** Unsupported builtin value: BaseVertex
Base vertex	OpenGL 4.5 (GLSL)	Extension	**naga:** ERROR: 0:14: 'gl_BaseVertex' : undeclared identifier **SPIRV-Cross:** needs GL_ARB_shader_draw_parameters
Base vertex	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** ERROR: 0:18: 'gl_BaseVertex' : undeclared identifier **SPIRV-Cross:** SPIRV-Cross threw an exception: BaseVertex not supported in ES profile.
Base instance	WebGPU (WGSL)	No	**naga:** Unsupported builtin value: BaseInstance

Concept	Backend	Status	Reason (tools)
Base instance	OpenGL 4.5 (GLSL)	Extension	**naga:** ERROR: 0:14: 'gl_BaseInstance' : undeclared identifier **SPIRV-Cross:** needs GL_ARB_shader_draw_parameters
Base instance	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** ERROR: 0:18: 'gl_BaseInstance' : undeclared identifier **SPIRV-Cross:** SPIRV-Cross threw an exception: BaseInstance not supported in ES profile.
Clip distance	WebGPU (WGSL)	Extension	**naga:** needs clip_distances
Clip distance	WebGL 2 (GLSL ES 3.00)	Extension	**naga:** needs GL_EXT_clip_cull_distance (unverified: the compiler does not implement the extension) **SPIRV-Cross:** needs GL_EXT_clip_cull_distance (unverified: the compiler does not implement the extension)
Render target layer	WebGPU (WGSL)	No	**naga:** Entry point main at Vertex is invalid: Struct member 1 is missing a binding
Render target layer	OpenGL 4.5 (GLSL)	Extension	**naga:** Entry point main at Vertex is invalid: Struct member 1 is missing a binding **SPIRV-Cross:** needs GL_ARB_shader_viewport_layer_array
Render target layer	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** Entry point main at Vertex is invalid: Struct member 1 is missing a binding **SPIRV-Cross:** needs GL_NV_viewport_array2
Viewport index	WebGPU (WGSL)	No	**naga:** Entry point main at Vertex is invalid: Struct member 1 is missing a binding
Viewport index	OpenGL 4.5 (GLSL)	Invalid	**naga:** Entry point main at Vertex is invalid: Struct member 1 is missing a binding **SPIRV-Cross:** ERROR: 0:8: 'gl_ViewportIndex' : required extension not requested: Possible extensions include: GL_ARB_shader_viewport_layer_array
Viewport index	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** Entry point main at Vertex is invalid: Struct member 1 is missing a binding **SPIRV-Cross:** ERROR: 0:8: 'gl_ViewportIndex' : undeclared identifier
Primitive index	WebGPU (WGSL)	Extension	**naga:** needs primitive_index
Primitive index	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** needs GL_OES_geometry_shader **SPIRV-Cross:** ERROR: 0:9: 'gl_PrimitiveID' : undeclared identifier
Stencil reference (output)	Vulkan (SPIR-V)	Extension	**spirv-val:** needs SPV_EXT_shader_stencil_export
Stencil reference (output)	Direct3D 12 (HLSL SM6)	No	**naga:** unsupported extension SPV_EXT_shader_stencil_export **SPIRV-Cross:** SPIRV-Cross threw an exception: Unsupported builtin in HLSL: 5014
Stencil reference (output)	WebGPU (WGSL)	No	**naga:** unsupported extension SPV_EXT_shader_stencil_export
Stencil reference (output)	OpenGL 4.5 (GLSL)	Extension	**naga:** unsupported extension SPV_EXT_shader_stencil_export **SPIRV-Cross:** needs GL_ARB_shader_stencil_export
Stencil reference (output)	WebGL 2 (GLSL ES 3.00)	No	**naga:** unsupported extension SPV_EXT_shader_stencil_export **SPIRV-Cross:** SPIRV-Cross threw an exception: Stencil export not supported in GLES.
Sample index	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** needs GL_OES_sample_variables **SPIRV-Cross:** needs GL_OES_sample_variables
Sample mask	WebGPU (WGSL)	No	**naga:** Global variable [2] '' is invalid: Initializer doesn't match the variable type
Sample mask	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** Global variable [2] '' is invalid: Initializer doesn't match the variable type **SPIRV-Cross:** needs GL_OES_sample_variables
Sample position	Direct3D 12 (HLSL SM6)	No	**naga:** unsupported builtin %19 **SPIRV-Cross:** SPIRV-Cross threw an exception: Unsupported builtin in HLSL: 19
Sample position	WebGPU (WGSL)	No	**naga:** unsupported builtin %19

Concept	Backend	Status	Reason (tools)
Sample position	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** unsupported builtin %19 **SPIRV-Cross:** needs GL_OES_sample_variables
Helper invocation	WebGPU (WGSL)	No	**naga:** unsupported builtin %23
Helper invocation	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** unsupported builtin %23 **SPIRV-Cross:** ERROR: 0:9: 'gl_HelperInvocation' : undeclared identifier
Barycentric coordinates	Vulkan (SPIR-V)	Extension	**spirv-val:** needs SPV_KHR_fragment_shader_barycentric
Barycentric coordinates	WebGPU (WGSL)	Invalid	**naga:** error: Entry point main at Fragment is invalid: Argument 0 varying error; Capability Capabilities(SHADER_BARYCENTRICS) is not supported (accepted by n
Barycentric coordinates	OpenGL 4.5 (GLSL)	Extension	**naga:** needs GL_EXT_fragment_shader_barycentric **SPIRV-Cross:** needs GL_EXT_fragment_shader_barycentric
Barycentric coordinates	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** needs GL_EXT_fragment_shader_barycentric **SPIRV-Cross:** SPIRV-Cross threw an exception: gl_BaryCoordEXT requires ESSL 320.
Barycentric coordinates (no perspective)	Vulkan (SPIR-V)	Extension	**spirv-val:** needs SPV_KHR_fragment_shader_barycentric
Barycentric coordinates (no perspective)	WebGPU (WGSL)	Invalid	**naga:** error: Entry point main at Fragment is invalid: Argument 0 varying error; Capability Capabilities(SHADER_BARYCENTRICS) is not supported (accepted by n
Barycentric coordinates (no perspective)	OpenGL 4.5 (GLSL)	Extension	**naga:** needs GL_EXT_fragment_shader_barycentric **SPIRV-Cross:** needs GL_EXT_fragment_shader_barycentric
Barycentric coordinates (no perspective)	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** needs GL_EXT_fragment_shader_barycentric **SPIRV-Cross:** SPIRV-Cross threw an exception: gl_BaryCoordNoPerspEXT requires ESSL 320.
Conservative depth	WebGPU (WGSL)	Lost	**naga:** output compiles but lacks /early_depth_test\(greater_equal\)/
Conservative depth	WebGL 2 (GLSL ES 3.00)	Lost	**naga:** needs GL_EXT_conservative_depth **SPIRV-Cross:** output compiles but lacks /depth_greater/
Global invocation id	WebGL 2 (GLSL ES 3.00)	No stage	—
Local invocation id	WebGL 2 (GLSL ES 3.00)	No stage	—
Local invocation index	WebGL 2 (GLSL ES 3.00)	No stage	—
Workgroup id	WebGL 2 (GLSL ES 3.00)	No stage	—
Number of workgroups	WebGL 2 (GLSL ES 3.00)	No stage	—
Workgroup size (built-in value)	WebGL 2 (GLSL ES 3.00)	No stage	—
Workgroup size (method attribute)	WebGL 2 (GLSL ES 3.00)	No stage	—
Subgroup size	WebGPU (WGSL)	Extension	**naga:** needs subgroups
Subgroup size	OpenGL 4.5 (GLSL)	Extension	**naga:** needs GL_ARB_compute_shader, GL_ARB_shader_storage_buffer_object **SPIRV-Cross:** needs GL_KHR_shader_subgroup_basic, GL_NV_shader_thread_group, GL_ARB_shader_int64, GL_ARB_shader_ballot, GL_AMD_gpu_shader_int64, GL_NV_gpu_shader5,
Subgroup size	WebGL 2 (GLSL ES 3.00)	No stage	—
Subgroup invocation id	WebGPU (WGSL)	Extension	**naga:** needs subgroups

Concept	Backend	Status	Reason (tools)
Subgroup invocation id	OpenGL 4.5 (GLSL)	Extension	**naga:** needs GL_ARB_compute_shader, GL_ARB_shader_storage_buffer_object **SPIRV-Cross:** needs GL_KHR_shader_subgroup_basic, GL_NV_shader_thread_group, GL_ARB_shader_int64, GL_ARB_shader_ballot
Subgroup invocation id	WebGL 2 (GLSL ES 3.00)	No stage	—
Subgroup id	WebGPU (WGSL)	Extension	**naga:** needs subgroups
Subgroup id	OpenGL 4.5 (GLSL)	Extension	**naga:** needs GL_ARB_compute_shader, GL_ARB_shader_storage_buffer_object **SPIRV-Cross:** needs GL_KHR_shader_subgroup_basic, GL_NV_shader_thread_group
Subgroup id	WebGL 2 (GLSL ES 3.00)	No stage	—
Number of subgroups	WebGPU (WGSL)	Extension	**naga:** needs subgroups
Number of subgroups	OpenGL 4.5 (GLSL)	Extension	**naga:** needs GL_ARB_compute_shader, GL_ARB_shader_storage_buffer_object **SPIRV-Cross:** needs GL_KHR_shader_subgroup_basic, GL_NV_shader_thread_group
Number of subgroups	WebGL 2 (GLSL ES 3.00)	No stage	—
Read-only storage buffer	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** The selected version doesn't support Features(BUFFER_STORAGE | DYNAMIC_ARRAY_SIZE) **SPIRV-Cross:** ERROR: 0:5: 'std430' : not supported for this version or the enabled extensions
Read/write storage buffer	WebGL 2 (GLSL ES 3.00)	Invalid	**naga:** The selected version doesn't support Features(BUFFER_STORAGE | DYNAMIC_ARRAY_SIZE) **SPIRV-Cross:** ERROR: 0:5: 'std430' : not supported for this version or the enabled extensions
64-bit float (double)	Metal (MSL)	No	**naga:** Type [2] '' is invalid: Using `f64` values requires the `naga::valid::Capabilities::FLOAT64` flag **SPIRV-Cross:** SPIRV-Cross threw an exception: double types are not supported in buffers in MSL.
64-bit float (double)	WebGPU (WGSL)	Invalid	**naga:** error: Type [0] '' is invalid: Using `f64` values requires the `naga::valid::Capabilities::FLOAT64` flag (accepted by naga only with non-WebGPU capabi
64-bit float (double)	WebGL 2 (GLSL ES 3.00)	No	**naga:** The selected version doesn't support Features(DOUBLE_TYPE) **SPIRV-Cross:** SPIRV-Cross threw an exception: FP64 not supported in ES profile.
Compute stage	WebGL 2 (GLSL ES 3.00)	No stage	—

Tools and versions

Tool	Version
`dxc`	libdxcompiler.dylib: 1.9(5399-a107ba61)(1.9.0.5399)
`naga`	30.0.1
`spirv-cross`	Git commit: vulkan-sdk-1.4.350.0-32-g6c09849f Timestamp: 2026-07-27T16:15:42
`glslang`	Glslang Version: 11:16.2.0
`spirv-tools`	SPIRV-Tools v2026.2 unknown hash, 2026-06-29T13:50:01+00:00
`metal`	Apple metal version 32023.921 (metalfe-32023.921.6)

Generated: 2026-10-17T21:06:58.474Z

How it was measured

Each concept is a small shader. HLSL shaders are compiled to SPIR-V with DXC, and the concepts HLSL cannot spell are written in Vulkan GLSL and compiled with glslang. Every SPIR-V module is validated with `spirv-val` (Vulkan 1.2) and then translated by two independent translators, **naga** and **SPIRV-Cross**, to HLSL (Shader Model 6), MSL, WGSL, GLSL 4.50 and GLSL ES 3.00. Each output is compiled by the backend's own tool: DXC for Direct3D 12, Apple's `metal` compiler for Metal, glslang for OpenGL and WebGL 2, and naga for WGSL. A concept is supported when at least one translator produces output that compiles.

Limits: Direct3D 11 (FXC, Shader Model 5) was not validated; no shader was run on a real device; results depend on the tool versions listed above. To regenerate the data, run `node tools/compat/run.mjs` in the website repository.

How the IDE uses this

The targets of a project are listed once in its `.csproj`:

```
<PropertyGroup>
  <YotsubaTargets>spirv;hlsl;msl;wgsl;glsl;glsles</YotsubaTargets>
</PropertyGroup>
```

Target	Backend	Diagnostic in the editor
`spirv`	Vulkan (SPIR-V)	Error for *No*, *Invalid* and *Lost*; warning for *Extension*.
`hlsl`	Direct3D 12 (HLSL SM6)	Same rules.
`msl`	Metal (MSL)	Same rules.
`wgsl`	WebGPU (WGSL)	Same rules.
`glsl`	OpenGL 4.5 (GLSL)	Same rules.
`glsles`	WebGL 2 (GLSL ES 3.00)	Same rules, plus an error for any stage marked *No stage*.

The rules are the same whatever syntax the shader is written in: `[SV_RenderTargetArrayIndex]`, `[gl_Layer]`, `[render_target_array_index]` and `[BuiltIn_Layer]` are one concept and get one diagnostic. Remove a target from `<YotsubaTargets>` and its diagnostics disappear.